Penn Engineering
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FA26

Special Seminar on
Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

https://cis7000-fall26.thelabone.org/

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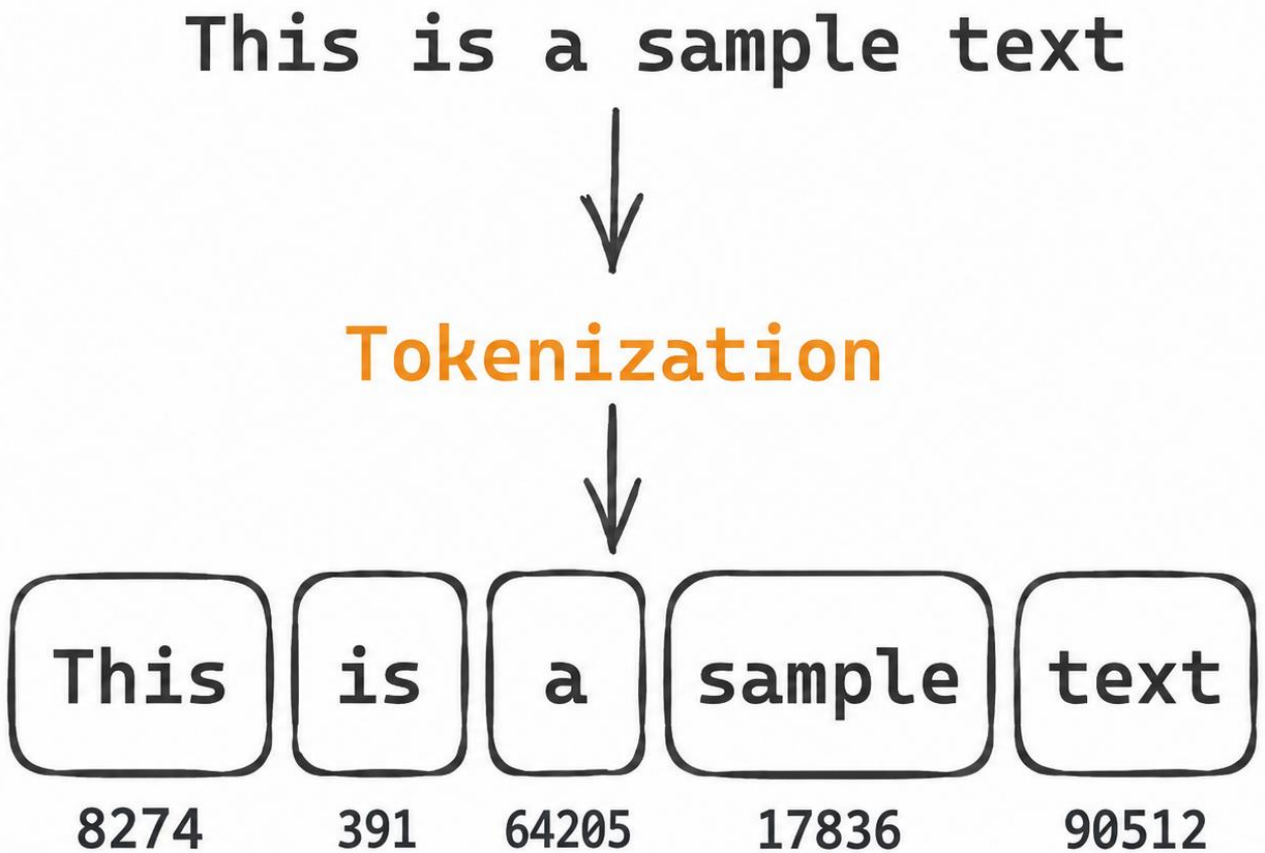
Lecture 12: Tokenization Across Modalities

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What is Tokenization



Why Tokenization

- Obtain a integer / continuous-vector sequence that encodes the input

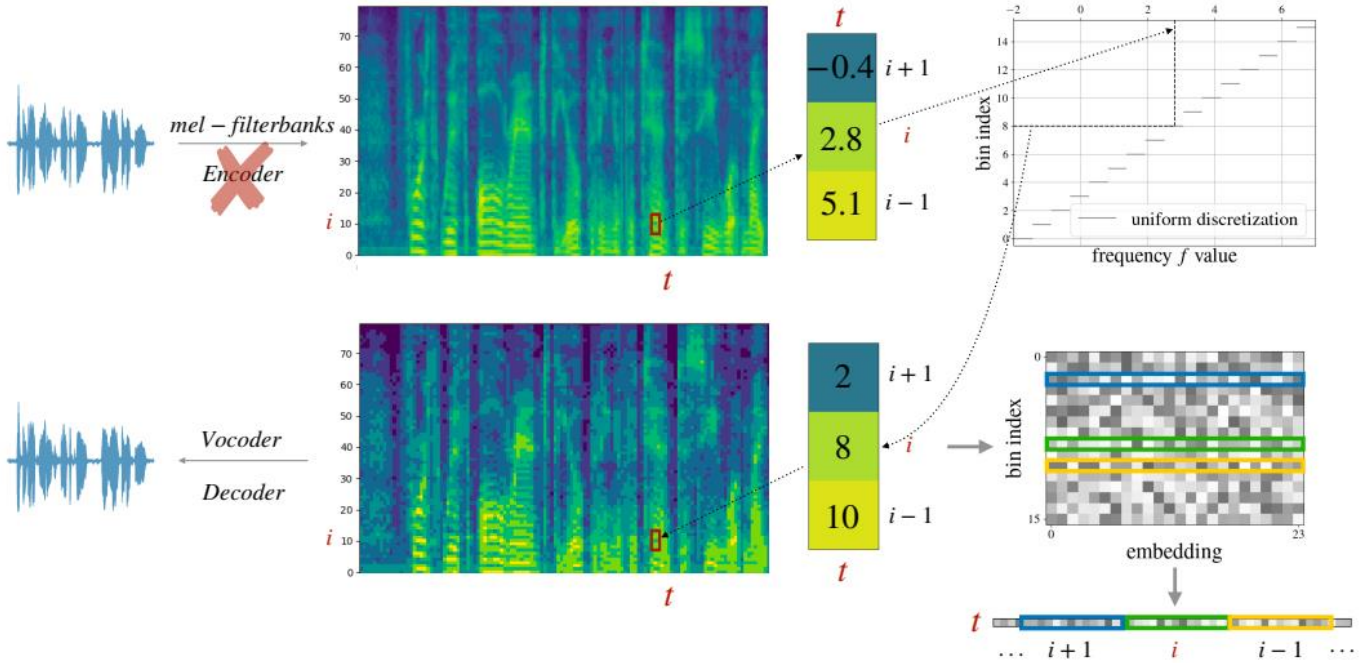
- Unified interface: model everything with a single model
  - After the encoders/tokenizers, everything is treated the same way
  - Easy to deploy
- Leverage synergy: if those tokens are aligned, or jointly learned

Ideas

- Leverage the signal structure
- Leverage the locality structures
- Multi-task learning

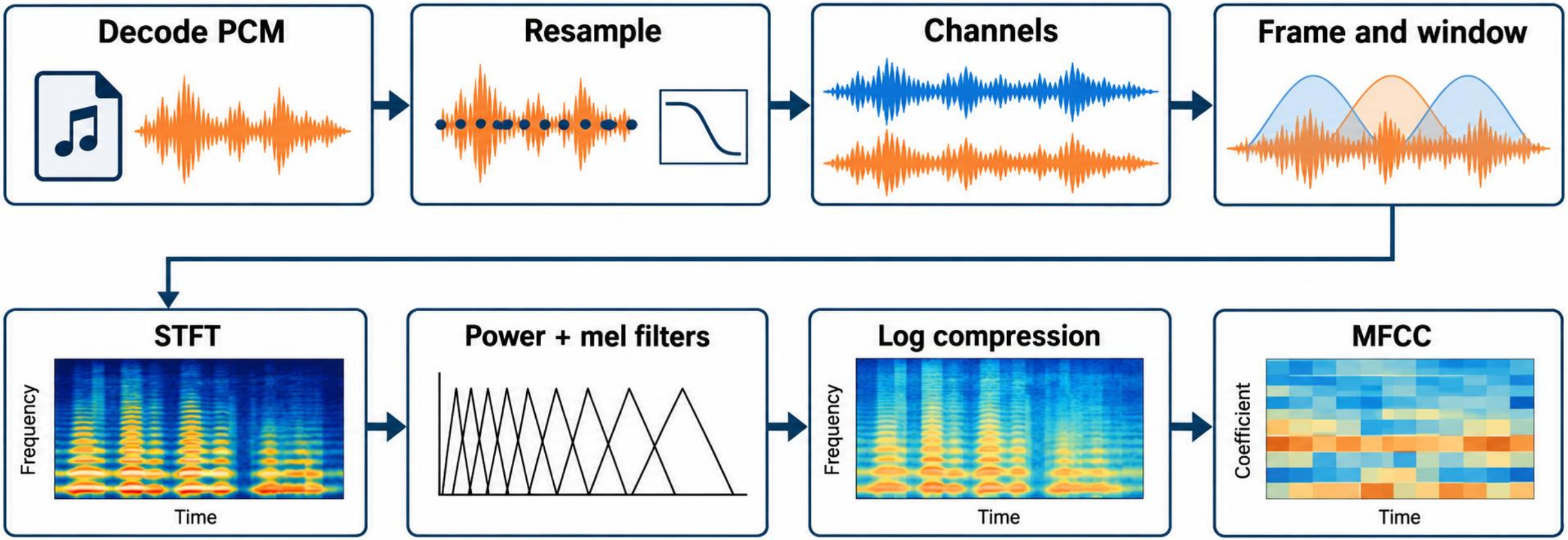
- More ideas on compression
- More ideas on ordering

dMel: direct signal quantization

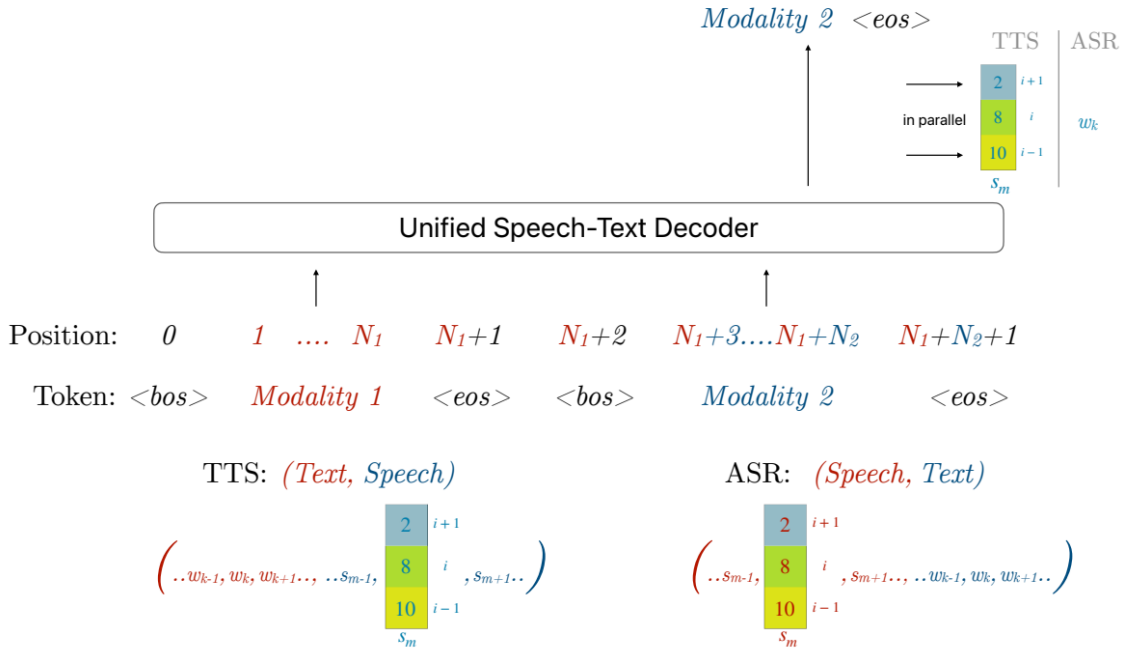


Bai et al., dMel (2024), Figure 1

Key idea: bin mel-filterbank energies directly to retain acoustic detail without learning an encoder


Decode PCM
Resample
Channels
Frame and window
STFT
Frequency
Time
Power + mel filters
Log compression
Frequency
Time
MFCC
Coefficient
Time

dMel: one time step contains many codes



Bai et al., dMel (2024), Figure 2

Channel embeddings concatenate and project to one frame embedding

TTS predicts all frequency channels in parallel at each autoregressive time step

dMel training and inference

Paired speech and transcripts train a decoder-only transformer with cross-entropy on the target modality

LibriSpeech supplies 960 hours for ASR, TTS, and joint experiments

Span masking reduces reliance on copying neighboring speech frames

At inference, a separate learned vocoder turns predicted bins into audio

dMel reconstruction evidence

Representation / vocoder	WER ↓	MOS-LQO ↑
Original speech	2.02	—
Mel / HiFiGAN	2.08	4.52
dMel / HiFiGAN	2.11	4.47
dMel / PWG, 40 Hz	2.51	4.29

300 LibriSpeech test-clean utterances, reconstruction evaluation

WER measures transcription errors; MOS-LQO estimates signal quality

Leverage the Signal Structure

A simple quantizer can retain information useful for both recognition and synthesis

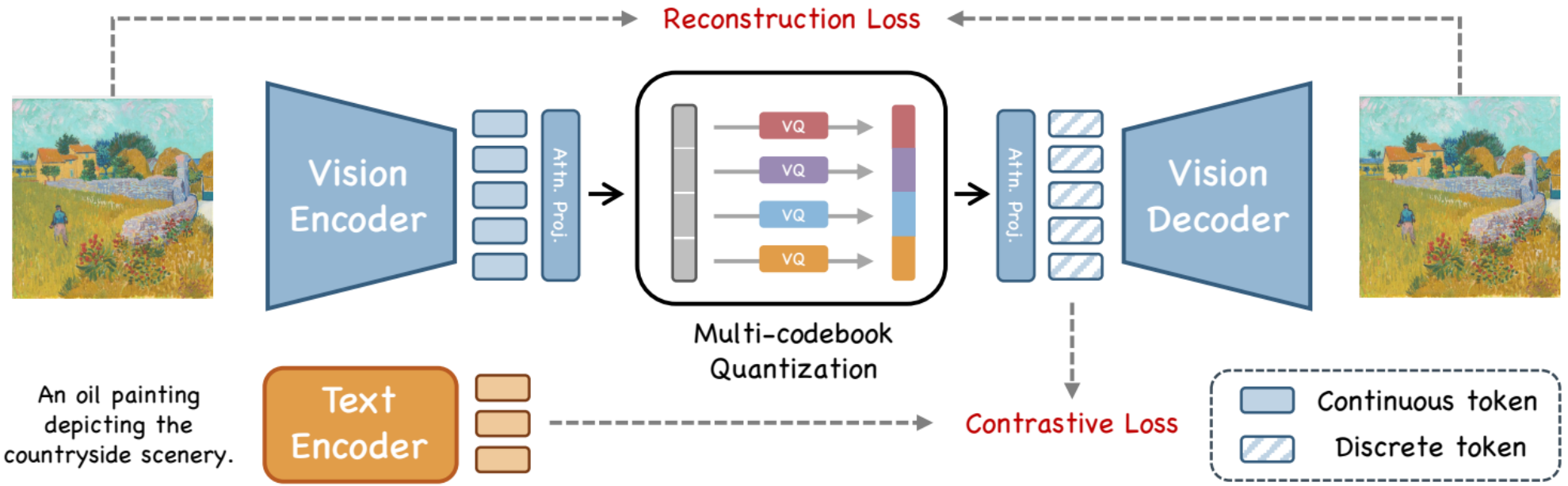
dMel trades a small vocabulary for many parallel codes in each "frame"

The Understanding-Generation Gap

Reconstruction asks whether the code retains the fine-grained details

Understanding asks whether the code supports the correct answer

UniTok: visual detail and semantics



Ma et al., UniTok (2025), Figure 2

Key idea: expand quantized feature capacity while jointly training reconstruction and image–text alignment

UniTok’s joint learning objective

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda \mathcal{L}_{\text{contra}}$$

More code capacity lets these signals coexist with less quantization loss

UniTok, Eqs. 1–2

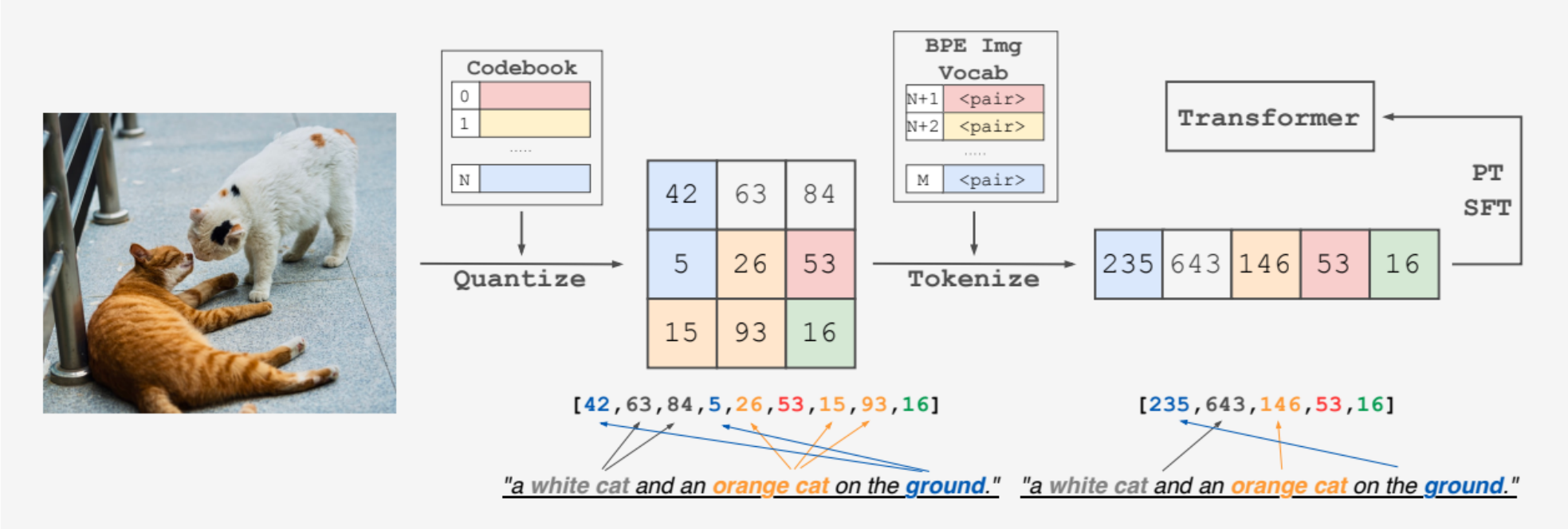
UniTok: locating the bottleneck

Ablation stage	Average VQA ↑
CLIP baseline	65.9
+ Channel factorization	59.8
+ Discretization	57.6
+ Reconstruction	57.5
+ Multi-codebook + attention projection	64.8

The largest drop occurs before reconstruction supervision is added

UniTok, Figure 3

Visual BPE: recurring spatial patterns



Zhang et al., From Pixels to Tokens (2024), Figure 1

Key idea: merge frequent adjacent VQ codes to model recurring spatial pattern

Horizontal and vertical adjacency are allowe

Visual BPE vocabulary and model training

2.78M images train merge rules over a pretrained VQ-GAN representation

Caption pretraining learns new image-token embeddings for Llama-3.1

1.27M LLaVA-OneVision examples support full supervised fine-tuning

At inference, fixed VQ and BPE rules convert an image into IDs for the language model

Visual BPE: matched-baseline results

PT(freeze) + SFT	VQAv2 ↑	MMBench ↑	POPE ↑
VQ only	55.4	37.6	76.0
VQ + BPE	57.1	40.9	79.0

Visual BPE, Table 5.1

Why merging can make learning easier

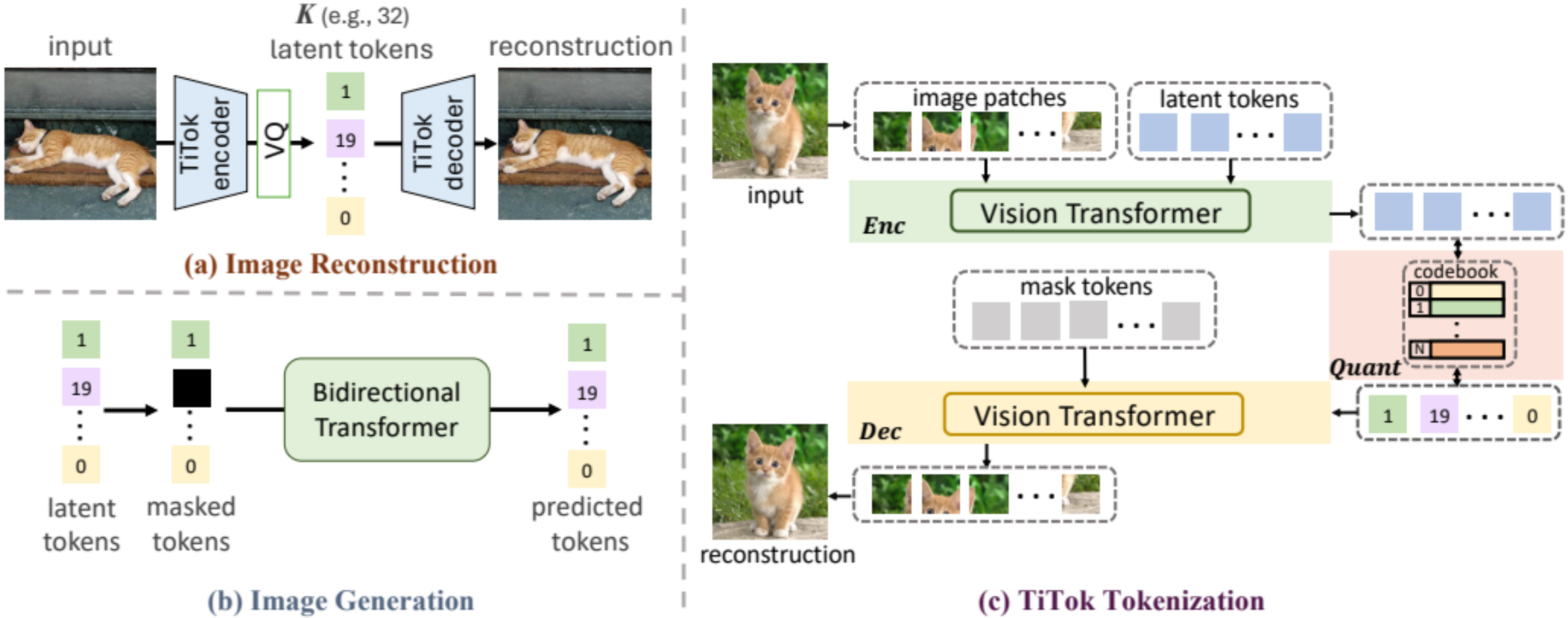
Merging dependent symbols can make a simple token model approach the source entropy rate (we remove low-entropy parts)

Intuition: the sequence should be “information-dense”

1D Tokenizers

Intuition: Let’s just compress more…

TiTok: a short global token sequence



Key idea: learned latent slots attend to all image patches, removing the fixed grid correspondence

TiTok: a short global token sequence

```
codes = init_latent_tokens()
_, codes = encoder_attention(concat(patchify(image), codes))

codes = quantize(codes)

patches = init_mask_tokens()
_, patches = decoder_attention(concat(codes, patches))
reconstruction = unpatchify(project(patches))
```

Key idea: learned latent slots attend to all image patches, removing the fixed grid correspondence

TiTok training and generation

Warm-up predicts proxy codes from an off-the-shelf MaskGIT VQ-GAN

Decoder fine-tuning freezes the encoder and quantizer

A separate MaskGIT model learns to predict randomly masked TiTok codes

TiTok reconstruction and generation

ImageNet 256 × 256	Tokens	rFID ↓	gFID ↓
MaskGIT VQ-GAN	256	2.28	6.18
TiTok-L-32	32	2.21	2.77
TiTok-B-64	64	1.70	2.48

All shown generators use MaskGIT-ViT with eight sampling steps

Tokenizer sizes differ, so this comparison does not isolate sequence length

Compact codes and decoder capacity

TiTok can summarize global scene structure with fewer latent slots

A larger tokenizer improves reconstruction when the slot budget is small

The mug’s identity may survive even when its exact surface texture does not

Limit: 32 tokens is an empirical operating point, not a universal bound

More Ideas about the Sequence: “Causality”

$$p(z_1, \ldots, z_L) = \prod_{i=1}^{L} p(z_i \mid z_{<i})$$

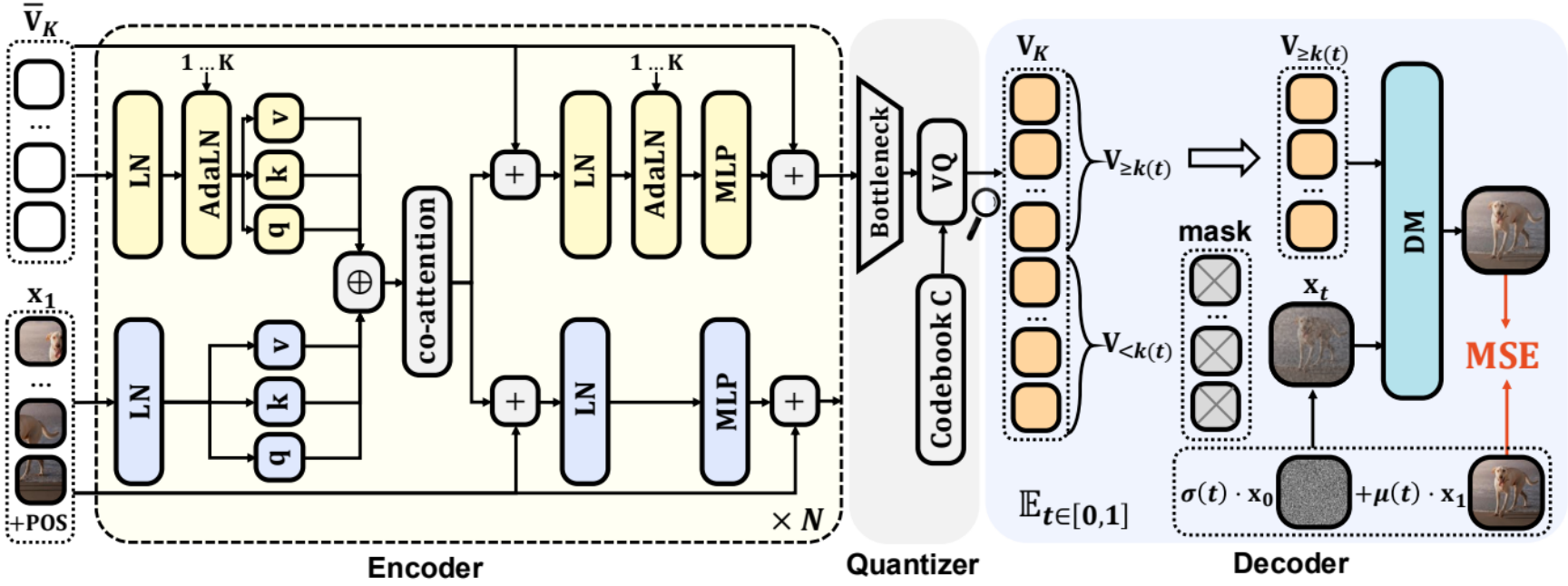
Any joint distribution admits a left-to-right chain-rule factorization

The order can still be easy or difficult for a finite model to learn

Selftok proposes an additional structural prior tied to a diffusion trajectory

General probability identity; Selftok §2

Selftok: learn tokens through denoising



Selftok Team (2025), Figure 7

Key idea: pair a noisy intermediate state with a token suffix so the suffix learns the remaining reconstruction

Intuition: the information in a prefix = the info. in a partial diffusion process

Selftok's reconstruction objective

$$x_t = (1 - t)x_0 + tx_1$$

$$\mathcal{L} = \mathrm{E}[\|x_1 - \mathrm{D}(x_t, V_{\geq k(t)})\|^2] + \mathcal{L}_Q$$

The clean image latent mixes with Gaussian noise at a sampled time

The decoder receives that noisy latent and the scheduled token suffix

Selftok, Eq. 8

Selftok’s token schedule

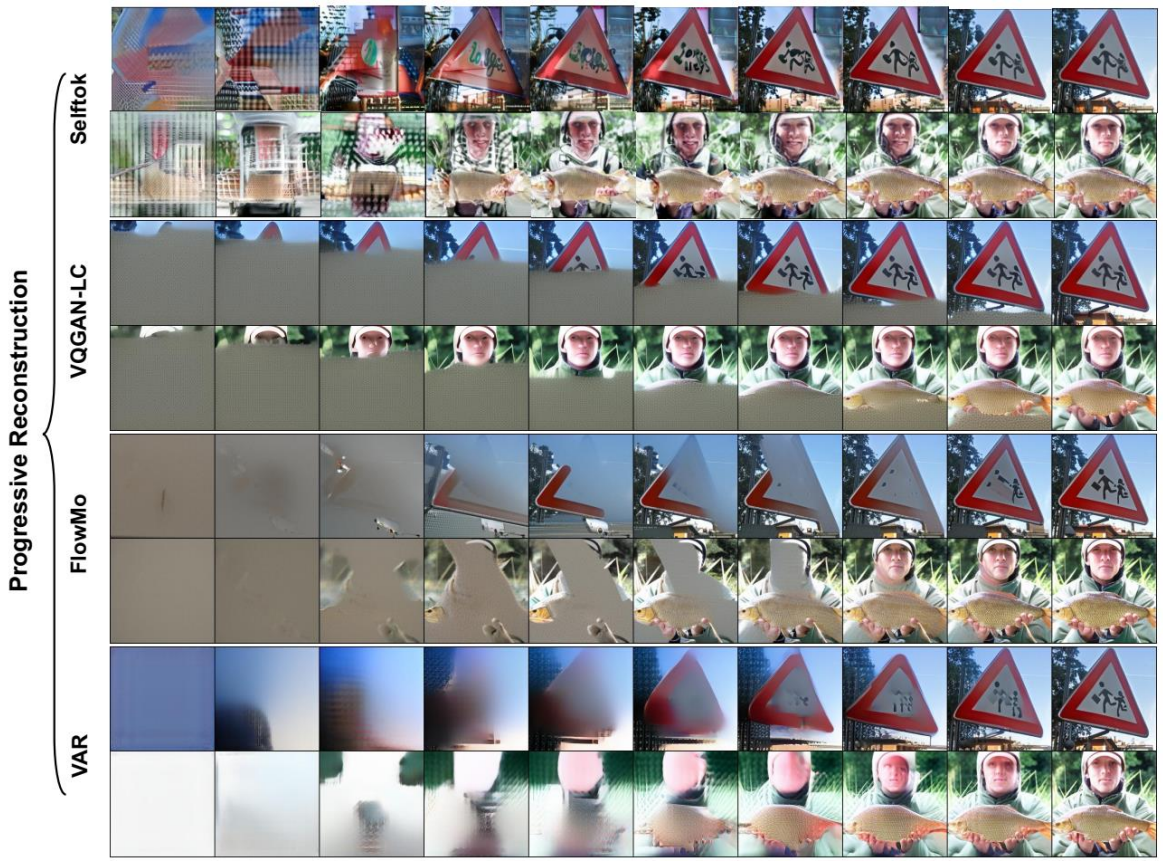
Near pure noise, the decoder receives most or all tokens

Near the clean image, fewer late tokens must explain the remaining detail

The paper reweights gradients because early tokens participate less often

Selftok, §§2.2.3–2.2.4 and Table 2

Selftok’s reconstruction progression



Selftok Team (2025), Figure 3, progressive reconstruction crop

The sequence changes global content instead of filling successive grid patches

Evidence for a different token order, not a guarantee of causal disentanglement

Selftok evidence and open claims

Mechanism: noise-conditioned suffix training ties tokens to a reconstruction progression

Evidence: reconstruction, ordering probes, downstream tasks, and reward-based improvements

Selftok, §§2–4; final question is teaching synthesis

Questions for Discussion

- What is a meaningful “ordering” structure for the domain you are interested in?

- Why do you think researchers have used VQ models to obtain discrete indices? Why not directly using continuous vectors if anyway we are going to “embed” them?